Group Division (code and report)
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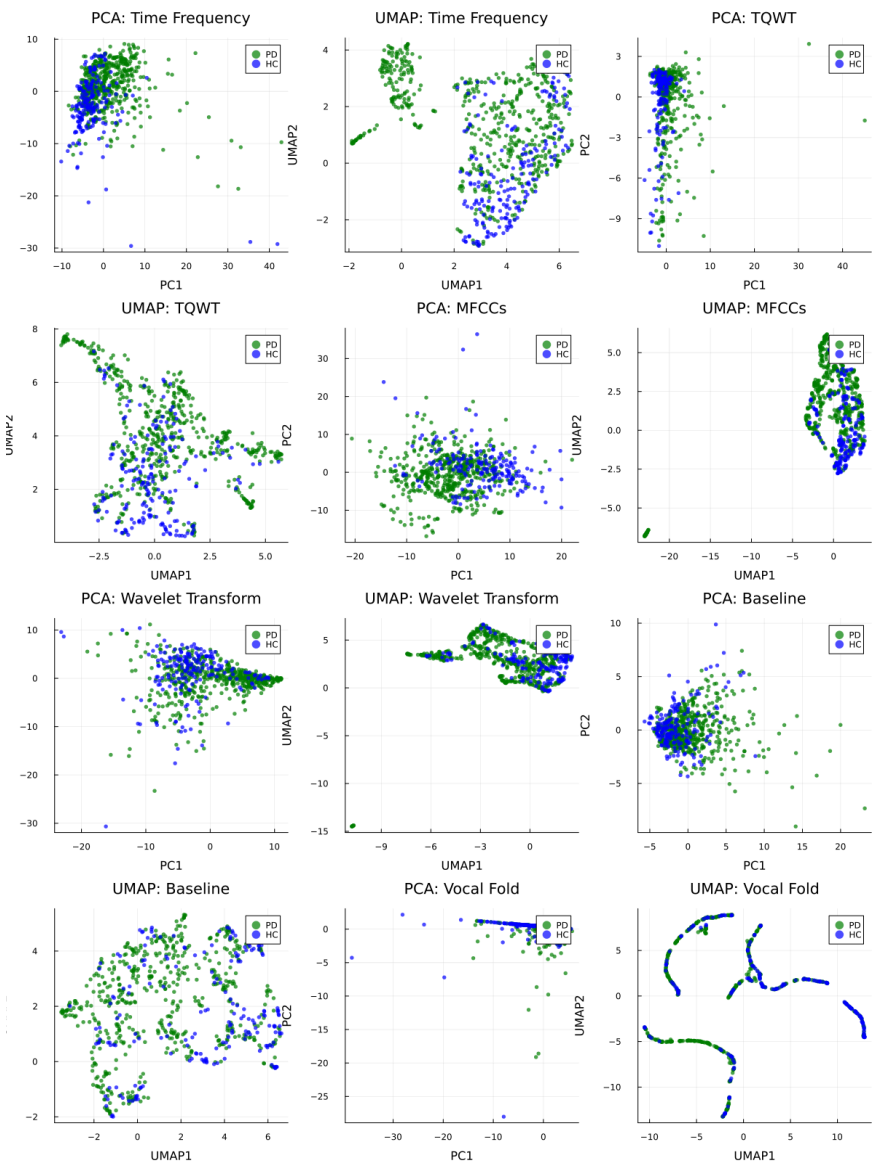
Task 1

Before constructing any models, we aimed to determine whether we could visually discern a difference between the PD and healthy control (HC) groups within each feature set. To achieve this, we employed two dimensionality reduction tools, PCA and UMAP, to transform the complex, high-dimensional data into straightforward 2D scatter plots.



We extracted six feature schemes from the PD voice dataset Baseline, Time-frequency, Vocal-fold, MFCCs, Wavelet, and TQWT and applied two unsupervised 2D visualization methods to each scheme to evaluate PD vs. HC separability. All repetitions were plotted with consistent styling (PD in blue, HC in green) to facilitate direct comparison across schemes.

Overall, we did not observe a clear two-dimensional separation: across both PCA and UMAP, PD and HC largely overlap, suggesting that simple linear or low-dimensional unsupervised structures do not effectively distinguish the classes. In terms of visual separability, the ranking is approximately MFCCs ≈ Time-frequency (weak–moderate) > Wavelet ≈ Vocal-fold ≈ Baseline > TQWT (weak). MFCCs and Time-frequency show the most potential, with UMAP revealing lobes/bands and slight PD–HC density shifts, whereas TQWT is the least promising (PCA compresses variance into a thin stripe; UMAP forms a horseshoe with significant mixing). Several schemes display curved ribbons/arches in UMAP—indicative of intrinsic manifolds (e.g., speaker/repeat/recording conditions) that do not align with PD labels—and instances where PCA concentrates variance (e.g., strong PC1 in TQWT) without enhancing discrimination. These patterns suggest limited performance from unsupervised/linear approaches; effective classification will likely require supervised learning (e.g., regularized linear models, kernels/boosting) and/or higher-dimensional representations (e.g., PCA-20→UMAP, deep embeddings) with subject-wise cross-validation to prevent leakage.

In conclusion, the MFCCs and TQWT feature schemes offer the clearest visual distinction between the PD (green) and HC (blue) clusters. In contrast, feature sets such as "Time Frequency" and "Baseline" exhibit significant overlap, while the "Vocal Fold" features show almost no separation, suggesting they might be less effective for classification.

Task 2

We evaluated the predictive effectiveness of each of the six feature schemes through a multi-step process derived from the code. Initially, we prepared the data and models by using three classifiers: k-Nearest Neighbors (kNN), Random Forest, and Logistic Regression. To ensure fairness in testing, we implemented a "group k-fold cross-validation." This method was essential as it keeps all voice recordings from a single person together—either for training the model or for testing it, but never both simultaneously. The code then iterated through each feature scheme, selected the relevant columns, and cleaned the data by removing any unusable features. For each cleaned feature set, the code cycled through all three classifiers, performing the 5-fold cross-validation and recording the accuracy for each fold.

Result

The results were compiled into a summary table to point out the best-performing combinations. For instance, when the Logistic Regression model was tested on the "Time Frequency" features, the code calculated a mean accuracy of 78.8% across the five folds. The complete summary is provided below.

Scheme	Classifier	Mean Accuracy
Wavelet Transform	**Random Forest**	**0.800**
TQWT	Random Forest	0.790
Time Frequency	Logistic Regression	0.788
Time Frequency	Random Forest	0.788
Baseline	Logistic Regression	0.787
MFCCs	Random Forest	0.784
MFCCs	Logistic Regression	0.783

Scheme	Classifier	Mean Accuracy
Wavelet Transform	Logistic Regression	0.783
TQWT	Logistic Regression	0.776
TQWT	kNN	0.771
Time Frequency	kNN	0.769
Baseline	Random Forest	0.766
Vocal Fold	Logistic Regression	0.752
Vocal Fold	Random Forest	0.750
MFCCs	kNN	0.745
Baseline	kNN	0.725
Vocal Fold	kNN	0.684

The table below presents a breakdown of each classifier's performance across all six feature schemes. It illustrates how effectively each model identified Healthy Controls (Class 0) and Parkinson's Disease patients (Class 1) overall.

Classifier	True Negative (TN)	False Positive (FP)	False Negative (FN)	True Positive (TP)	Overall Accuracy
Random Forest	412	740	**259**	**3125**	**78.0%**
Logistic Regression	**426**	**726**	280	3104	77.8%
kNN	361	791	359	3025	74.6%

- TN (True Negative): Healthy individuals correctly identified as healthy.
- FP (False Positive): Healthy individuals incorrectly identified as having PD.
- FN (False Negative): PD patients incorrectly identified as healthy.
- TP (True Positive): PD patients correctly identified as having PD.

Analysis

The Random Forest model achieved the highest overall accuracy at 78.0%. It excelled in identifying PD patients, recording the most True Positives (3125) and the fewest missed diagnoses (259 False Negatives). The Logistic Regression model slightly outperformed in identifying healthy individuals, with the highest number of True Negatives (426). A notable weakness across all models was their difficulty in identifying healthy controls (Class 0). This is evidenced by their low Recall scores. This indicates that the models more frequently misclassified healthy individuals as having PD than they missed actual PD cases.

Task 3

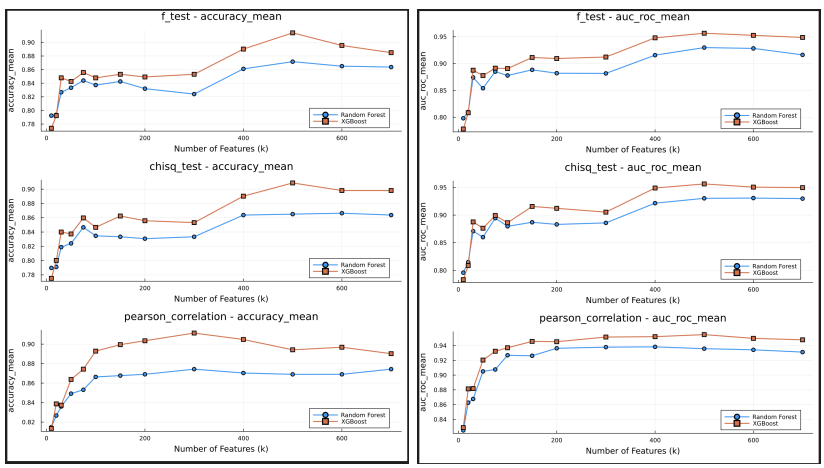
A. Methods

a. Three filter-based feature selection methods were compared:
   i. F-test: Measures statistical significance of linear relationships between individual features and the class label.
   ii. Pearson Correlation: Quantifies linear correlation between features and target variable.
   iii. Chi-square Test: Assesses dependency between categorical variables

b. Also we pick two ensemble-based approaches:
   i. XGBoost: Gradient boosting
   ii. Random Forest: Bagging-based method for comparison

c. Optimization Strategy

A three phase rough-to-fine parameter optimization was implemented with phase 1 (rough search) used sample of *k = [10, 20, 30, 50, 75, 100, 150, 200, 300, 400, 500, 600, 700]* and also used 5-fold cross validation for initial performance screening.



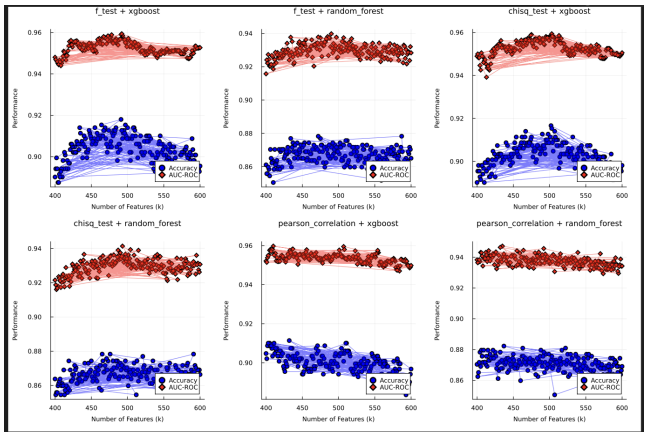
Next, in the 2nd phase, we focused on the most promising ranges (k=400-600) based on our finding in the first phase with step size = 1 and compared it with 3,5,7.10 folds. Lastly, we identify the optimal configurations under both Accuracy and AUC-ROC.

B. Key Findings and Results

Metric	Filter Method	Number of Features	Classifiier	Accuracy	AUC-ROC
Best by Accuracy	F-test	491	XGBoost	0.918	0.959

Best by AUC-ROC	Pearson Correlation	410	XGBoost	0.9087	0.9597

The F-test model maximized classification accuracy, while the Pearson correlation model got superior AUC-ROC. As for the full result graph it can be seen in the following.



Feature Importance Analysis

Jitter Measures, Shimmer Measures, Format Frequencies seems to be a high importance factor in this due to repeating features saw in the results.

Optimization Insights

XGBoost always outperformed RandomForest due to its ability to model non-linear relationships.

F-test optimized for high accuracy, Pearson correlation is for superior AUC performance.

C. Discussion

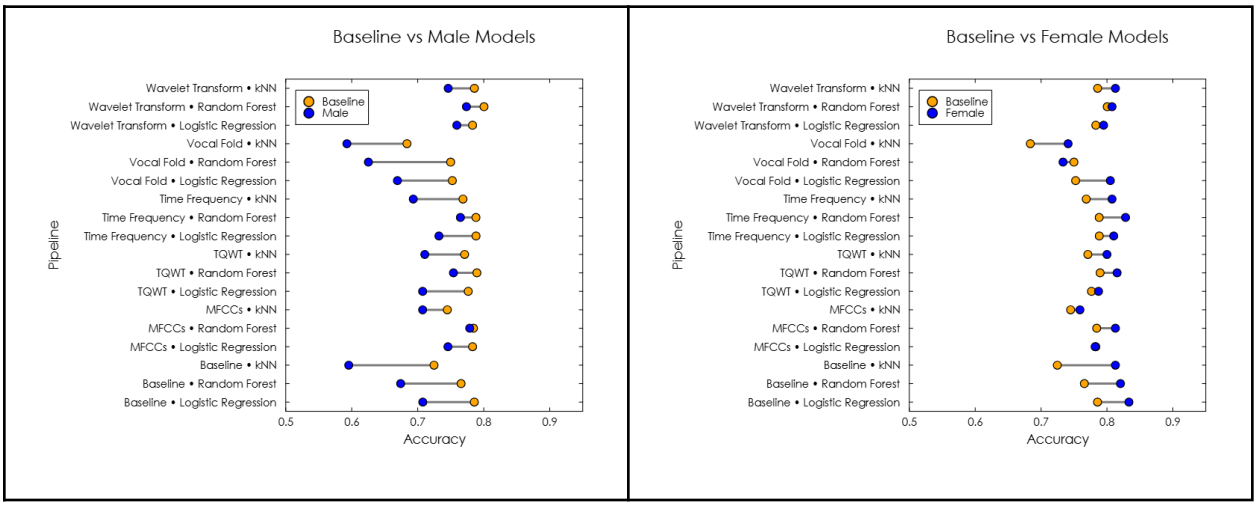
If we prioritize raw classification accuracy, we would pick F-test with 491 features but the fact that f-test used more features than its contestant Pearson-correlation showed that its sacrifice computational cost for higher accuracy which in this case is not a lot. But, what if we prioritize overall discriminative ability, Pearson correlation with 410 features would be a better choice due to how well AUC-ROC measures. By the concept, F-test only selects features based on variance explained between classes, which favors statistically significant whereas pearson correlation selects features with highest linear correlation to the target which helps remove redundancy and help cut down features that are just redundant.

Task 4

A. Gender-specific models

Male and female voices can have different frequency patterns, which may cause one model to fit better for one gender. Thus, we trained separate models for male and female samples using the same features and cross-validation folds.

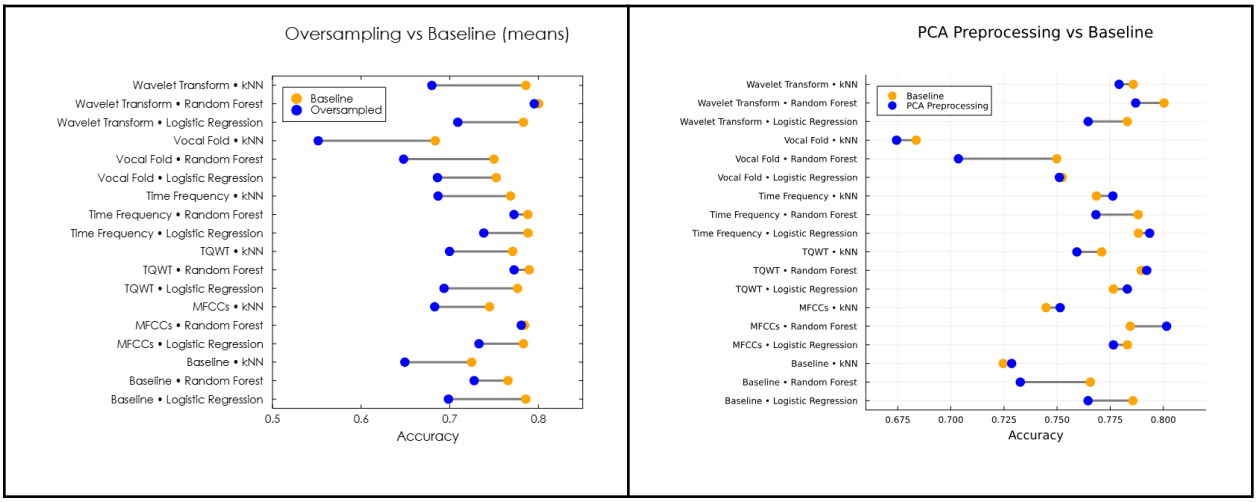
The female models performed slightly better than the baseline, while the male models mostly performed worse. This may be because the female samples are more consistent, while the male data is more diverse, which reduces the model's accuracy.



B. Addressing class imbalance and noises

a. Oversampling

Minority samples (PD) in the training set were duplicated within each fold to create a balanced distribution. Overall, this version performed worse than the baseline. The repetitive samples might make the model overfit and less generalizable.

b. PCA Preprocessing

We standardized features and applied PCA (95% variance) to reduce noise before training. Some models performed slightly better than the baseline, particularly when using MFCC and TQWT features.